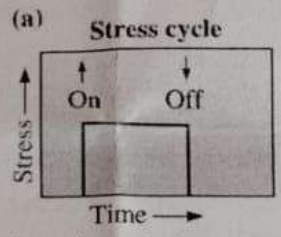
End Semester (Even Semester) Examination –2025-26

Subject: ***Materials Science and Engineering (MO-201)***

Time 3 hours ***Full marks 60***

1. Draw strain - time response for the given stress cycle for elastomer, viscoelastic material, and viscous materials **(2 marks).**



2. Draw the engineering stress and engineering strain curve for metals, and label the elastic limit, offset yield strength, tensile strength, breaking strength, and strain to failure. **(2 marks)**
3. Steel and aluminium have Young's moduli of 210 GPa and 70 GPa, respectively. Tensile samples of steel and aluminium are made with identical geometry. In both samples, the yield stress occurs above 40,000 psi. At an engineering stress of 30,000 psi, the steel sample shows an elastic strain of 0.001. Find the elastic strain of the aluminium sample at an engineering stress of 30,000 psi. **(2 marks)**
4. Materials response to external force changes depending on temperature, loading, and strain rate. Name the material behavior phenomenon that led to the failure of
   a) Titanic in 1912, b) World War II Liberty ship, c) Space Shuttle Challenger in 1986, and d) Space Shuttle Columbia in 2003 **(2 marks)**
5. What is Poisson's ratio, and comment on its value for metals in the elastic region and the plastic region. **(2 marks)**
6. What is slip and slip system? **(2 marks)**
7. Can shear stress parallel to tensile stress result in the yielding of a single crystal? Give justification **(2 marks)**
8. Name two modes through which plastic deformation can take place in metallic samples. Give at least two key differences between the two modes. **(2 marks)**
9. Explain the grain boundary strengthening mechanism and the associated relation with neat schematics **(4 marks)**
10. Show that the probability of occupancy of energy level *E* by an electron is (i) negligible for E >> EF, (ii) 0.5 for E=EF, (iii) ~1 for E<< EF. **(2 marks)**
11. The Fermi level for potassium is 2.1 eV. Calculate the velocity of the electrons at the Fermi level. **(2 marks)**
12. Explain why nichrome and not copper is used as a heating element. **(1 mark)**
13. Explain why aluminium used in long distance transmission lines cannot be strengthened by solid solution. **(1 mark)**
14. Explain why lead and zinc with an even number of electrons in the outer shell and a full valence band are conductors. **(2 marks)**
15. Why does the conductivity of a semiconductor change with impurity content? Compare this with the behaviour of metallic conductors. **(2 marks)**
16. The resistance of a sample of pure germanium has the following temperature dependence:

Temperature, K	312	354	385	420
Resistance, ohm	11.8	2.33	0.90	0.35

Determine the energy gap in germanium. **(2 marks)**

17. A CdTe photoconductor responds to a glowing cigarette. Explain why this is so. **(1 mark)**

18. Determine the wavelength of the light from a GaAs laser, assuming that the radiative transition occurs between the bottom of the conduction band and the top of the valence band. **(2 marks)**

19. The dielectric constant of polyethylene is independent of temperature, while that of polyvinylchloride is not. Explain this difference in behaviour on the basis of their monomer structures. **(2 marks)**

20. When ice melts into water, the dielectric constant increases, in contrast to the decrease observed during the melting of HCl. Explain why this is so. **(1 mark)**

21. What is the variation expected in the loss factor in a sodium silicate glass, when some of the Na+ ions are replaced by Rb ions? Explain. **(2 marks)**

22. Calculate the polarization of a $BaTiO_3$ crystal. The shift of the titanium ion from the body centre is 0.06 Å. The oxygen anions of the side faces shift by 0.06 Å, while the oxygen anions of the top and bottom faces shift by 0.08 Å, all in a direction opposite to that of the titanium ion. **(2 marks)**

23. Does a large magnetic moment per atom mean a high Curie temperature? Explain. **(1 mark)**

24. Explain how a high initial permeability in Fe–Ni alloys helps to reduce the area under the hysteresis loop. **(2 marks).**

25. The total iron loss in a transformer core at normal flux density was measured at 25 Hz and at 50 Hz and was found to be 250 W and 800 W respectively. Calculate (i) the eddy current loss and (ii) the hysteresis loss at 50 Hz. **(2 marks)**

26. The electrical resistivity of iron increases by a factor of 6, on alloying with 4% silicon. Estimate the decrease in eddy current losses for this change in composition. **(2 marks)**

27. In the 1st transition metal element series only Fe, Co, Ni are ferromagnetic. Explain this observation from exchange energy viewpoint. **(2 marks)**

28. None of the constituting elements in Heusler alloys such as Cu_2MnSn and Cu_2MnAl are ferromagnetic yet they show ferromagnetic behaviour. Give reason for this observation. **(1 mark)**

29. In nickel ferrite, $NiFe_2O_4$, the ferric ions are antiferromagnetically coupled. The magnetization is due to the nickel ions. When zinc is added to nickel ferrite, the magnetization of the crystal increases, even though the zinc ions are not ferromagnetic. Explain how this could happen.**(2marks)**

30. d) Fe-4% Si is a preferred material for transformer core whereas Ni-Zn ferrites or other oxide magnetic materials are used for high frequency applications like microwave isolators. Explain the reason behind this choice from energy loss aspect for soft magnetic materials. **(2 marks)**

31. Draw and explain conductivity vs temperature plot for an extrinsic semiconductor. **(2 marks)**

32. Why GaAs is preferred over Si in optoelectronic communication? **(1 mark)**